

International Trade - Assignment 2

Theory questions, summarizing Eaton, Kortum & Sotelo (2012):

a What is the challenge they try to overcome?

Most international trade models with firm heterogeneity use a continuum of firms, following the approach of Dornbusch, Fischer, and Samuelson (1977). This method, widely adopted in the literature, is used by Melitz (2003), Chaney (2008), and Eaton and Kortum (2002) to simplify analyses by rendering each firm negligible at the aggregate level.

However, the authors raise several problems with this continuum hypothesis:

- **Individual firm shocks not affecting aggregates:** In continuum models, each firm has a zero measure, meaning that none can influence aggregates, such as the price index, making it impossible to reflect the impact of individual fluctuations on aggregate data.
- **Explanation of zero (or small) trade flows:** In continuum models, a zero trade flow implies that market conditions were such that no company could theoretically export. Indeed, in Helpman, Melitz and Rubinstein (2008), this is achieved by truncating the continuous Pareto distribution to a certain efficiency support. Less efficient companies are excluded, which may lead to the absence of exports to certain countries. In reality, it could simply be a case where no company found it profitable to export, even if it was possible to do so. Continuum models therefore ignore individual variations or strategic decisions that can naturally lead to zero flows without imposing an insurmountable barrier.
- **Restrictions on parameter values to obtain finite solutions:** Continuum models require strict restrictions on parameters to obtain finite solutions (such as $\theta > \sigma - 1$, θ being the shape parameter of the Pareto distribution used to model the efficiency of firms, and σ the elasticity of substitution between goods), which prevents company size and sales distributions from being too skewed. Empirical observations show, however, that these distributions may be highly skewed.

Therefore, the authors propose an innovative trade model with firm heterogeneity, this time with a **finite number of firms** rather than a continuum, which allows fluctuations affecting individual companies to be reflected in the aggregates; predicts zero flows when, by chance, no company is competitive enough to export to a given country; and eliminates the need to restrict certain distribution parameters to obtain finite values.

The authors mention that the idea of using a finite-firm model to explain zeros flows had already been proposed by Armenter and Koren (2008), but their approach was "purely probabilistic" rather than economic (treating trade outcomes as a matter of statistical probability without relying on the economic decisions and entry costs of firms).

b Describe briefly the model they describe.

The international trade model by Eaton, Kortum, and Sotelo (2012) builds on heterogeneous firm models by treating firms as a finite set rather than a continuum. Here's a deeper look into the key components and the mechanics of the model:

Firm efficiency and Pareto distribution

Firm efficiencies Z are assumed to be drawn from a Pareto distribution with shape parameter θ and lower bound $z_0 > 0$. For a $z \geq z_0$, the probability that a firm's efficiency is greater or equal than z is given by:

$$P(Z \geq z) = \left(\frac{z_0}{z}\right)^\theta$$

Instead of assuming, as in the continuum case, that the quantity of firms that have efficiency $Z \geq z$ in a country i is exactly $T_i z^{-\theta}$, with T_i being the overall measure of firms in the country (also reflecting the level of innovation), the actual number of firms with efficiency greater than z is modeled here as a Poisson random variable, with parameter $T_i z^{-\theta}$. The quantity of firms is therefore no longer deterministic, which introduces randomness in trade flows.

Cost of exporting

Each firm faces a cost C_{ni} to export from country i to country n . This cost depends on the wage in the origin country, w_i , the trade cost factor d_{ni} (equal to 1 for domestic sales), and the firm's efficiency Z :

$$C_{ni} = \frac{w_i d_{ni}}{Z}$$

Only firms with costs low enough to make a profit enter a foreign market.

Price setting and market entry:

Gross profit in market n of a firm with unit cost C is:

$$\Pi_n(C) = \left[1 - \frac{C}{p_n(C)}\right] \left(\frac{p_n(C)}{P_n}\right)^{-(\sigma-1)} X_n,$$

with P_n the price index and X_n the total spending in country n .

By ordering firm costs from lowest to highest, and denoting K_n the number of firms that enter the market in country n , the price index is:

$$P_n^{K_n} = \left(\sum_{k=1}^{K_n} \left[p_n^{K_n} \left(C_n^{(k)} \right) \right]^{-(\sigma-1)} \right)^{-1/(\sigma-1)}$$

As this is a finite sum, the price index is affected by individual price variation (due to a cost variation coming from a productivity shock for instance), therefore firm's pricing decision depends directly on the prices set by other firms (Bertrand competition).

K_n is determined by the following entry conditions:

$$\Pi_n^{K_n} \left(C_n^{(K_n)} \right) \geq E_n \quad \text{and} \quad \Pi_n^{K_n+1} \left(C_n^{(K_n+1)} \right) < E_n.$$

with E_n the entry cost.

Unlike continuum models, there is no parameter restriction needed to obtain finite solutions and it is possible that no firm enters the market simply due to a lack of luck, as none are able to afford the entry cost.

c Explain how their gravity model differs from the one in Eaton & Kortum (2002).

General differences between the two models

First of all, EK (2002) uses a continuum of firms, whereas EKS (2012), as detailed above, builds a model with a finite number of firms, thus allowing randomness in firm entry.

Secondly, an important difference is that EK (2002) models technology variation at the national level, whereas EKS (2012) incorporates firm-level heterogeneity within countries. In EK (2002), each country's efficiency is Fréchet-distributed, while in EKS (2012), firm-level efficiency follows a Pareto distribution.

Moreover, in EK (2002), intermediate goods plays a big role in amplifying trade flows and increasing sensitivity to trade costs (notably through the β parameter). EKS (2012) abstracts from this structure, achieving trade variability through the discrete nature of firm entries rather than through intermediate input linkages.

Also, the price index in EK (2002) takes the form of an integral involving the Gamma function $\Gamma\left(\frac{\theta+1-\sigma}{\theta}\right)$, which requires the restriction $\theta > \sigma - 1$ to be well defined (with θ the Fréchet distribution parameter and σ the elasticity of substitution), whereas the price index defined as a finite sum in EKS (2012) involves no particular restriction. Individual firm-level fluctuations are also not reflected in EK (2002), whereas they are in EKS (2012).

Finally, it would appear that the bilateral trade data used in EK (2002) does not contain zero trade flows, as the model they develop is specifically designed to explain positive trade flows among countries, whereas zero trade flows are addressed in EKS (2012).

Differences in the gravity equations

The two gravity equations in themselves are significantly different in the sense that the EK (2002) equation involves the cost of intermediate goods, while the EKS (2012) equation involves the wage in country i . They also differ because EKS (2012) involves expected trade shares, whereas EK (2002) involves actual trade shares (using a deterministic approach).

$$\frac{X_{ni}}{X_n} = \frac{T_i(c_i d_{ni})^{-\theta}}{\sum_{k=1}^N T_k(c_k d_{nk})^{-\theta}} ; \quad E \left[\frac{X_{ni}}{X_n} \right] = \frac{T_i(w_i d_{ni})^{-\theta}}{\sum_{k=1}^N T_k(w_k d_{nk})^{-\theta}}$$

EK (2002) EKS (2012)

with c_i the input cost, w_i the wage, T_i the state of technology and d_{ni} the trade cost factor.

The way these two equations are estimated are also quite different, on the one hand because EK (2002) uses self-imports to estimate the gravity equation (i.e. normalizes trade share equation by "self-imports" share in n), whereas this is not the case for EKS (2012). This leads to the following adjusted gravity equation for EK (2002):

$$\ln \frac{X'_{ni}}{X'_{nn}} = S_i - S_n - \theta d_k - \theta b - \theta l - \theta e_n - \theta m_n - \theta \delta_{ni}$$

with $\ln X'_{ni} = \ln X_{ni} - \left[\frac{(1-\beta)}{\beta} \right] \ln \left(\frac{X_i}{X_{ii}} \right)$ and δ_{ni} the error term.

Whereas EKS (2012) presents the following gravity equation suitable for estimation:

$$E \left[\frac{X_{ni}}{X_n} \right] = \frac{\varphi_{ni} \nu_{ni}}{\sum_l \varphi_{nl} \nu_{nl}}.$$

with

$$\varphi_{ni} = \begin{cases} \exp(S_i + m_n + g'_{ni} \alpha) & i \neq n \\ \exp(S_n) & i = n \end{cases},$$

$$-\theta \ln d_{ni} = m_n + g'_{ni} \alpha + \ln \nu_{ni},$$

$$S_i = \ln T_i - \theta \ln w_i,$$

and ν_{ni} the error term.

On the other hand, the estimators they use are quite different:

- EK (2002) uses the Generalized Least Squares (GLS) estimator,
- EKS (2012) uses the Pseudo Maximum Likelihood (PML) one.

d Describe their counterfactual experiments.

In the Eaton, Kortum, and Sotelo (2012) model, the authors perform two main counterfactual experiments:

- **Reducing Trade Barriers by 10% Globally:** They simulate a 10% decrease in trade barriers worldwide. This reduction in barriers leads to the introduction of 206 new bilateral trading relationships. However, these new trade flows account for only a very small quantity of the trade increase induced by the barrier reduction. The global impact leads to +43% exports worldwide (99.9984% of which is between already trading countries). This experiment is designed to examine the impact of small changes in trade barriers on the structure of international trade networks, especially on the establishment of new trade links.

- **Resampling Firm Efficiencies:** They resample the efficiencies of individual firms globally to study the variability in aggregate outcomes, such as the manufacturing price index. They find that this resampling generates a significant variance in price indices across countries. For instance, the manufacturing price index for the United States shows a 14% variance, while Denmark's index varies by 24%. This experiment addresses the implications of firm-level randomness on macroeconomic aggregates, challenging the zero variance implied by models with a continuum of firms.

International Trade - Assignment 2

2nd part - Empirical questions

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I'll do the next part of the assignment with the R software, displaying my code, tables and comments for each question below.

```
library(haven)
library(dplyr)
library(tidyr)
library(knitr)
library(fixest)
library(broom)
library(ggplot2)
library(didimputation)
```

2. Provide descriptive statistics on international trade (drop “imports from self”)

a. share of (i) bilateral zero trade dyads and (ii) uni-directional zero trade dyads – both over time.

```
# Read the .dta file
data <- read_dta("DataForAssignment.dta")

data_without_Xii <- data %>%
  filter((exporter != importer))

data_without_Xii <- data_without_Xii %>%
  mutate(has_trade = ifelse(trade > 0, 1, 0))

dyad_summary <- data_without_Xii %>%
  select(year, exporter, importer, has_trade) %>%
  left_join(data_without_Xii %>%
    select(year, exporter = importer, importer = exporter, has_trade) %>%
    rename(has_trade_reverse = has_trade),
    by = c("year", "exporter", "importer")) %>%
  replace_na(list(has_trade_reverse = 0)) %>%
  mutate(
    bilateral_zero_trade = ifelse(has_trade == 0 & has_trade_reverse == 0, 1, 0),
    uni_directional_zero_trade = ifelse(xor(has_trade == 1, has_trade_reverse == 1), 1, 0)
  )

yearly_totals <- dyad_summary %>%
  group_by(year) %>%
  summarize(
    total_dyads = n(),
    bilateral_zero_trade_dyads = sum(bilateral_zero_trade),
    uni_directional_zero_trade_dyads = sum(uni_directional_zero_trade)
  ) %>%
  mutate(
    bilateral_zero_trade_percentage = (bilateral_zero_trade_dyads / total_dyads) * 100,
    uni_directional_zero_trade_percentage = (uni_directional_zero_trade_dyads / total_dyads) * 100
  )
```

```
output <- yearly_totals %>%
  select(year, bilateral_zero_trade_percentage, uni_directional_zero_trade_percentage)

kable(output, caption = "Share of bilateral zero trade
and uni-directional zero trade dyads", format = "markdown")
```

Table 1: Share of bilateral zero trade and uni-directional zero trade dyads

year	bilateral_zero_trade_percentage	uni_directional_zero_trade_percentage
1986	11.7220801	12.318841
1987	10.6990622	13.043478
1988	11.9352089	11.082694
1989	9.3350384	12.745098
1990	7.2463768	11.807332
1991	6.3938619	10.912191
1992	6.3086104	8.354646
1993	6.5217391	7.459505
1994	5.6265985	6.947997
1995	3.7510656	7.928389
1996	3.4953112	7.672634
1997	2.9411765	6.990622
1998	2.0034101	7.331628
1999	2.0460358	6.180733
2000	1.6197783	5.456095
2001	1.5771526	5.711850
2002	1.3213981	4.859335
2003	1.1935209	4.603581
2004	0.8951407	4.646206
2005	1.0230179	4.006820
2006	0.8951407	4.092072

b. share of countries linked by RTAs, over time.

```
rta_shares <- data_without_Xii %>%
  group_by(year) %>%
  summarize(
    total_dyads = n(),                # Total dyads for each year
    rta_dyads = sum(RTA)              # Dyads with RTA link
  ) %>%
  mutate(
    rta_share_percentage = (rta_dyads / total_dyads) * 100 # Calculate share as a percentage
  ) %>%
  select(year, rta_share_percentage) # Keep only relevant columns

kable(rta_shares, caption = "Share of countries linked by RTAs", format = "markdown")
```

Table 2: Share of countries linked by RTAs

year	rta_share_percentage
1986	0.000000
1987	0.000000
1988	0.000000
1989	3.239557
1990	4.859335
1991	5.242967
1992	8.610401
1993	10.272805
1994	11.082694
1995	11.849957
1996	11.849957
1997	14.535379
1998	15.132140
1999	15.558397
2000	16.922421
2001	17.092924
2002	18.115942
2003	18.968457
2004	21.227622
2005	21.696505
2006	22.037511

c. overall international trade (export + import) as ratio to GDP, over time.

```
#Overall international trade (export + import) as ratio to GDP, over time.
trade_by_year <- data_without_Xii %>%
  group_by(year) %>%
  summarise(total_trade_year = sum(trade, na.rm = TRUE))

gdp_by_year <- data_without_Xii %>%
  group_by(year) %>%
  distinct(exporter, .keep_all = TRUE) %>%
  summarise(total_gdp = sum(GDPexp, na.rm = TRUE))

trade_gdp_ratio <- trade_by_year %>%
  inner_join(gdp_by_year, by = "year") %>%
  mutate(trade_to_gdp_percentage = total_trade_year / total_gdp * 100) %>%
  select(year, trade_to_gdp_percentage)

kable(trade_gdp_ratio,
      caption = "Overall international trade as ratio to GDP (%)",
      format = "markdown")
```

Table 3: Overall international trade as ratio to GDP (%)

year	trade_to_gdp_percentage
1986	9.239408
1987	9.634449
1988	10.143379
1989	10.604168
1990	10.930850
1991	10.711608
1992	10.875543
1993	10.412690
1994	11.266855
1995	12.102603
1996	12.292450
1997	12.894155
1998	13.002950
1999	13.037891
2000	13.896632
2001	13.503434
2002	13.514596
2003	13.868538
2004	14.852089
2005	15.260524
2006	16.134916

The ratio is increasing every year. Note, however, that not every country in the world is included in the dataset (there are only 69 countries included).

Estimate the gravity model. - Although “internal distance” is defined for domestic trade (exporter==importer), other correlates are not: contiguity, common language, colonial ties, and being part of an RTA. Those are set to zero. Therefore, you need to always add a dummy for domestic trade. Otherwise, you will get nonsensical results.

3. Use data only for 1990 Estimate the gravity model with distance, contiguity, common language, colonial ties, and being part of an RTA as explanatory variables (be sure to add a dummy for domestic trade). Describe how the coefficients change across specifications, and why. Collect all these results in one table.

Code

a. Naïve gravity model without fixed effects, using importer and exporter GDPs, in logs, OLS.

```
data_1990_for_gravity_model <- data %>%
  filter(year == 1990) %>%
  mutate(log_distance = log(DIST),
         log_importer_gdp = log(GDPimp),
         log_exporter_gdp = log(GDPexp),
         domestic = ifelse(importer == exporter, 1, 0)
  )

data_1990_wo_zero <- data_1990_for_gravity_model %>%
  filter(trade!=0) %>%
  mutate(log_trade = log(trade))

ols_model <- feols(log_trade ~ log_distance + log_importer_gdp +
                  log_exporter_gdp + CNTG + LANG + CLNY + RTA + domestic,
                  data = data_1990_wo_zero, vcov=~pair_id)
# We cluster errors by pair_id
# to account for potential correlation in the error terms within each pair
```

b. Standard model with exporter and importer fixed effects, in logs, OLS.

```
fe_model <- feols(log_trade ~ log_distance + CNTG + LANG + CLNY + RTA + domestic
                 | importer + exporter,
                 data = data_1990_wo_zero, vcov=~pair_id)
```

c. Standard model with exporter and importer fixed effects, Poisson, using all observations, including all possible zeros.

```
# Using PPML
model_poisson <- fepois(
  trade ~ log_distance + CNTG + LANG + CLNY + RTA + domestic | importer + exporter,
  data = data_1990_for_gravity_model,
  vcov = ~pair_id
)
# We cluster errors by pair_id
# to account for potential correlation in the error terms within each pair
```

d. Standard model with exporter and importer fixed effects, Poisson, using all observations, including all possible zeros, using import shares as dependent variable (a la Sotelo).

```
# Calculation of import shares
data_1990_for_gravity_model <- data_1990_for_gravity_model %>%
  group_by(importer) %>%
  mutate(total_imports = sum(trade)) %>%
  ungroup()
```



```
data_1990_for_gravity_model <- data_1990_for_gravity_model %>%
  mutate(import_share = trade / total_imports)

# PPML
model_poisson_import_shares <- fepois(
  import_share ~ log_distance + CNTG + LANG + CLNY + RTA + domestic | importer + exporter,
  data = data_1990_for_gravity_model,
  vcov = ~pair_id
)
```

e. Like d, but restricting sample to strictly positive flows.

```
data_1990_wo_zero_w_import_shares <- data_1990_for_gravity_model %>%
  filter(trade!=0)

model_poisson_import_shares2 <- fepois(
  import_share ~ log_distance + CNTG + LANG + CLNY + RTA + domestic | importer + exporter,
  data = data_1990_wo_zero_w_import_shares,
  vcov = ~pair_id
)
```

f. Like d, for a sample without domestic trade.

```
data_1990_wo_domestic_w_import_shares <- data_1990_for_gravity_model %>%
  filter((exporter != importer))

model_poisson_import_shares3 <- fepois(
  import_share ~ log_distance + CNTG + LANG + CLNY + RTA | importer + exporter,
  data = data_1990_wo_domestic_w_import_shares,
  vcov = ~pair_id
)
```

g. Like d, but drop the dummy for domestic trade.

```
model_poisson_import_shares4 <- fepois(
  import_share ~ log_distance + CNTG + LANG + CLNY + RTA | importer + exporter,
  data = data_1990_for_gravity_model,
  vcov = ~pair_id
)
```

Overall results and comments Here is the code to display the full table.

```
# Collecting all model results
ols_model_results <- tidy(ols_model)
fe_model_results <- tidy(fe_model)
model_poisson_results <- tidy(model_poisson)
model_poisson_import_shares_results <- tidy(model_poisson_import_shares)
model_poisson_import_shares2_results <- tidy(model_poisson_import_shares2)
model_poisson_import_shares3_results <- tidy(model_poisson_import_shares3)
model_poisson_import_shares4_results <- tidy(model_poisson_import_shares4)

# Defining a formatting function (with coeff, standard error, p-value stars)
format_regression_table <- function(data, column_name = "formatted") {
  data$stars <- cut(data$p.value,
    breaks = c(-Inf, 0.001, 0.01, 0.05, 0.1, Inf),
    labels = c("***", "**", "*", ".", ""))
}
```

```

data[[column_name]] <- paste0(format(data$estimate, scientific = TRUE, digits = 2),
                              " (", format(data$std.error, scientific = TRUE,
                                             digits = 2), ") ",
                              data$stars)

result_table <- data[, c("term", column_name), drop = FALSE]

return(result_table)
}

# We format model results in a nice way, each time adding the number of observations
# and renaming variables in order to better combine them afterwards

# For OLS, log trade as dependent variable, no fixed effects
model_name = 'Model a'
ols_model_results <- format_regression_table(ols_model_results, column_name = model_name)
ols_model_results <- rbind(ols_model_results,
                          c("Number of observations", length(ols_model$fitted.values)))
ols_model_results$term <- gsub("log_importer_gdp", "log importer_gdp", ols_model_results$term)
ols_model_results$term <- gsub("log_exporter_gdp", "log exporter_gdp", ols_model_results$term)

# For OLS, log trade as dependent variable, exporter and importer fixed effects,
model_name = 'Model b'
fe_model_results <- format_regression_table(fe_model_results, column_name = model_name)
fe_model_results <- rbind(fe_model_results,
                          c("Number of observations", length(fe_model$fitted.values)))

# For Poisson, trade as dependent variable, exporter and importer fixed effects
model_name = 'Model c'
model_poisson_results <- format_regression_table(model_poisson_results, column_name = model_name)
model_poisson_results <- rbind(model_poisson_results,
                              c("Number of observations", length(model_poisson$fitted.values)))

# For Poisson, import share as dependent variable, exporter and importer fixed effects
model_name = 'Model d'
model_poisson_import_shares_results <-
  format_regression_table(model_poisson_import_shares_results, column_name = model_name)
model_poisson_import_shares_results <-
  rbind(model_poisson_import_shares_results,
        c("Number of observations", length(model_poisson_import_shares$fitted.values)))

# For Poisson, import share as dependent variable,
# exporter and importer fixed effects, only positive flows
model_name = 'Model e'
model_poisson_import_shares2_results <-
  format_regression_table(model_poisson_import_shares2_results, column_name = model_name)
model_poisson_import_shares2_results <-
  rbind(model_poisson_import_shares2_results,
        c("Number of observations", length(model_poisson_import_shares2$fitted.values)))

# For Poisson, import share as dependent variable
# exporter and importer fixed effects, without domestic trade
model_name = 'Model f'

```

```

model_poisson_import_shares3_results <-
  format_regression_table(model_poisson_import_shares3_results, column_name = model_name)
model_poisson_import_shares3_results <-
  rbind(model_poisson_import_shares3_results,
        c("Number of observations", length(model_poisson_import_shares3$fitted.values)))

# For Poisson, import shares as dependent variable,
# exporter and importer fixed effects, without dummy for domestic
model_name = 'Model g'
model_poisson_import_shares4_results <-
  format_regression_table(model_poisson_import_shares4_results, column_name = model_name)
model_poisson_import_shares4_results <-
  rbind(model_poisson_import_shares4_results,
        c("Number of observations", length(model_poisson_import_shares4$fitted.values)))

# Combining results
combined_results <-merge(ols_model_results, fe_model_results,
                        by = "term", all = TRUE)
combined_results <-merge(combined_results, model_poisson_results,
                        by = "term", all = TRUE)
combined_results <-merge(combined_results, model_poisson_import_shares_results,
                        by = "term", all = TRUE)
combined_results <-merge(combined_results, model_poisson_import_shares2_results,
                        by = "term", all = TRUE)
combined_results <-merge(combined_results, model_poisson_import_shares3_results,
                        by = "term", all = TRUE)
combined_results <-merge(combined_results, model_poisson_import_shares4_results,
                        by = "term", all = TRUE)

desired_order <- c("(Intercept)", "log exporter_gdp", "log importer_gdp", "log_distance", "CNTG", "LANG",
                  "CLNY", "RTA", "domestic", "Number of observations")
combined_results$term <- factor(combined_results$term, levels = desired_order)
combined_results <- combined_results[order(combined_results$term), ]
rownames(combined_results) <- NULL

```

Please find the full table on the next page.

```
kable(combined_results,
      caption = "Gravity Models Estimations for 1990",
      format = "markdown")
```

Table 4: Gravity Models Estimations for 1990

term	Model a	Model b	Model c	Model d	Model e	Model f	Model g
(Intercept)	-1.2e+01 (4.3e-01) ***	NA	NA	NA	NA	NA	NA
log ex- porter_gdp	1.2e+00 (1.5e-02) ***	NA	NA	NA	NA	NA	NA
log im- porter_gdp	8.9e-01 (1.5e-02) ***	NA	NA	NA	NA	NA	NA
log_distance	-9.9e-01 (3.8e-02) ***	-1.1e+00 (4.9e-02) ***	-6.7e-01 (9.9e-02) ***	-6.8e-01 (8.0e-02) ***	-6.5e-01 (8.0e-02) ***	-9.6e-01 (5.3e-02) ***	-2.1e+00 (4.9e-02) ***
CNTG	6.3e-01 (2.1e-01) **	3.9e-01 (2.1e-01) .	7.4e-01 (2.2e-01) ***	8.7e-01 (2.0e-01) ***	8.7e-01 (1.9e-01) ***	4.2e-01 (1.4e-01) **	-1.4e+00 (1.4e-01) ***
LANG	6.7e-01 (1.1e-01) ***	5.6e-01 (1.1e-01) ***	2.5e-01 (1.3e-01) .	6.1e-01 (1.2e-01) ***	5.9e-01 (1.2e-01) ***	4.3e-01 (1.0e-01) ***	1.7e-01 (1.4e-01)
CLNY	6.2e-01 (1.8e-01) ***	8.3e-01 (1.7e-01) ***	1.3e-01 (1.5e-01)	6.5e-01 (1.5e-01) ***	6.4e-01 (1.4e-01) ***	7.2e-01 (1.3e-01) ***	6.5e-01 (2.1e-01) **
RTA	1.9e-01 (1.9e-01)	-3.3e-01 (1.5e-01) *	2.1e-01 (1.3e-01)	4.8e-01 (2.8e-01) .	4.9e-01 (2.8e-01) .	-3.0e-01 (1.9e-01)	6.2e-02 (4.4e-01)
domestic	4.8e+00 (2.5e-01) ***	4.4e+00 (3.4e-01) ***	3.3e+00 (2.8e-01) ***	4.3e+00 (2.4e-01) ***	4.3e+00 (2.4e-01) ***	NA	NA
Number of observa- tions	4144	4144	4761	4761	4144	4692	4761

Each cell displays the estimated coefficient, followed by the standard error in parentheses. Asterisks indicate statistical significance: *** for $p < 0.001$, ** for $p < 0.01$, * $p < 0.05$ and . for $p < 0.10$

Model a: OLS, log trade as dependent variable, no fixed effects, only positive flows

Model b: OLS, log trade as dependent variable, exporter and importer fixed effects, only positive flows

Model c: Poisson, trade as dependent variable, exporter and importer fixed effects

Model d: Poisson, import share as dependent variable, exporter and importer fixed effects

Model e: Poisson, import share as dependent variable, exporter and importer fixed effects, only positive flows

Model f: Poisson, import share as dependent variable, exporter and importer fixed effects, without domestic trade

Model g: Poisson, import share as dependent variable, exporter and importer fixed effects, without dummy for domestic trade

Comment on differences across specifications in the estimates of the coefficients to distance and RTA. What do you conclude?

We can see that the effect of distance on trade (or import shares) is always negative and significant. This consistency in the result demonstrates the robustness of the negative relationship between trade (or import shares) and distance. However, we note that the magnitude of the distance coefficient varies slightly between models. This is due to variations in estimation techniques across models, such as whether GDP is included, the presence of a domestic dummy, the use of fixed effects, sample size reductions, etc.

Furthermore, we observe that the effect of RTA on trade is negative and significant for model b, which doesn't make much sense, as these types of agreements are expected to enhance trade. However, using the PPML estimator à la Sotelo, with import shares as the dependent variable (model d), we find a positive and quite significant effect of RTA on trade, which seems more plausible. This result holds when dropping zero trade flows, but it doesn't when dropping domestic trade or dropping the dummy for domestic trade: it results in a non-significant effect, negative for model f and positive for model g.

In conclusion, for our data from 1990, the best-suited regression model appears to be the Sotelo model, Poisson (PPML), with import shares as the dependent variable, using all observations, including all possible zeros (i.e., model d). This model provides a significant negative relationship between import shares and distance, and a positive one between import shares and RTA. It also shows positive and significant relationships between import shares and contiguity, common language, and colonial ties.

However, it appears challenging to estimate the effect of RTA on trade using only data from 1990, as results vary widely between models. Data including the time dimension for each importer-exporter pair might help us better estimate this effect.

4. Now estimate the following specification in panel, using all available years Include in the regression exporter*importer FEs, exporter*year FEs, importer*year FEs. You can only identify the time-varying effect of joining an RTA.

- a. do this three times, using a domestic trade dummy
 - by OLS (in logs)

```
data_panel_for_gravity_model <- data %>%
  mutate(log_distance = log(DIST),
         domestic = ifelse(importer == exporter, 1, 0),
         )

# By OLS (in logs)
data_panel_wo_zero <- data_panel_for_gravity_model %>%
  filter(trade!=0) %>%
  mutate(log_trade = log(trade))

# We cluster standard errors per country pairs
ols_model_rta <- feols(log_trade ~ RTA | importer^exporter
                      + importer^year + exporter^year,
                      data = data_panel_wo_zero, vcov = ~pair_id)
```

- by PPML

```
# By PPML
ppml_model_rta <- fepois(trade ~ RTA | importer^exporter
                        + importer^year
                        + exporter^year,
                        data = data_panel_for_gravity_model,
                        vcov = ~pair_id)
```

- by PPML-Sotelo

```
# By PPML - Sotelo (using import shares as the dependent variable)
# Calculation of import shares
data_panel_for_gravity_model <- data_panel_for_gravity_model %>%
  group_by(importer) %>%
  mutate(total_imports = sum(trade)) %>%
  ungroup()

data_panel_for_gravity_model <- data_panel_for_gravity_model %>%
  mutate(import_share = trade / total_imports)

# PPML - Sotelo
ppml_sotelo_model_rta <- fepois(import_share ~ RTA | importer^exporter
                               + importer^year + exporter^year,
  data = data_panel_for_gravity_model,
  vcov = ~pair_id)
```

b. do this three times, using domestic trade*year FEs.

- by OLS (in logs)

```
data_panel_for_gravity_model$coupleDomesticYear <-
  paste(data_panel_for_gravity_model$domestic,
        data_panel_for_gravity_model$year, sep = " ")

ols_model_rta_with_domYearFE <- feols(log_trade ~ RTA | importer^exporter
                                       + importer^year + exporter^year
                                       + domestic^year,
  data = data_panel_wo_zero, vcov = ~pair_id)
```

- by PPML

```
ppml_model_rta_with_domYearFE <- fepois(trade ~ RTA | importer^exporter
                                         + importer^year + exporter^year
                                         + domestic^year,
  data = data_panel_for_gravity_model,
  vcov = ~pair_id)
```

- by PPML-Sotelo

```
ppml_sotelo_model_rta_with_domYearFE <- fepois(import_share ~ RTA | importer^year
                                                + importer^year + exporter^year
                                                + domestic^year,
  data = data_panel_for_gravity_model,
  vcov = ~pair_id)
```

Overall results and comments Here is the code to display the full table.

```
ols_model_rta_results <- tidy(ols_model_rta)
ols_model_rta_with_domYearFE_results <- tidy(ols_model_rta_with_domYearFE)
ppml_model_rta_results <- tidy(ppml_model_rta)
ppml_model_rta_with_domYearFE_results <- tidy(ols_model_rta_with_domYearFE)
ppml_sotelo_model_rta_results <- tidy(ppml_sotelo_model_rta)
ppml_sotelo_model_rta_with_domYearFE_results <- tidy(ols_model_rta_with_domYearFE)
```

```

# For OLS model
model_name = 'OLS a.'
ols_model_rta_results <- format_regression_table(ols_model_rta_results, column_name = model_name)
ols_model_rta_results <- rbind(ols_model_rta_results,
                              c("Number of observations",length(ols_model_rta$fitted.values)))
ols_model_rta_results <- rbind(ols_model_rta_results,
                              c("dom*year FEs","No"))

model_name = 'OLS b.'
ols_model_rta_with_domYearFE_results <-
  format_regression_table(ols_model_rta_with_domYearFE_results, column_name = model_name)
ols_model_rta_with_domYearFE_results <- rbind(ols_model_rta_with_domYearFE_results,
                                              c("Number of observations",length(ols_model_rta_with_domYearFE$fitted.values)))
ols_model_rta_with_domYearFE_results <- rbind(ols_model_rta_with_domYearFE_results,
                                              c("dom*year FEs","Yes"))

# For PPML model
model_name = 'PPML a.'
ppml_model_rta_results <-
  format_regression_table(ppml_model_rta_results, column_name = model_name)
ppml_model_rta_results <- rbind(ppml_model_rta_results,
                              c("Number of observations",length(ppml_model_rta$fitted.values)))
ppml_model_rta_results <- rbind(ppml_model_rta_results,
                              c("dom*year FEs","No"))

model_name = 'PPML b.'
ppml_model_rta_with_domYearFE_results <-
  format_regression_table(ppml_model_rta_with_domYearFE_results, column_name = model_name)
ppml_model_rta_with_domYearFE_results <- rbind(ppml_model_rta_with_domYearFE_results,
                                              c("Number of observations",length(ppml_model_rta_with_domYearFE$fitted.values)))
ppml_model_rta_with_domYearFE_results <- rbind(ppml_model_rta_with_domYearFE_results,
                                              c("dom*year FEs","Yes"))

# For PPML-Sotelo model
model_name = 'Sotelo a.'
ppml_sotelo_model_rta_results <-
  format_regression_table(ppml_sotelo_model_rta_results, column_name = model_name)
ppml_sotelo_model_rta_results <- rbind(ppml_sotelo_model_rta_results,
                                      c("Number of observations",length(ppml_sotelo_model_rta$fitted.values)))
ppml_sotelo_model_rta_results <- rbind(ppml_sotelo_model_rta_results,
                                      c("dom*year FEs","No"))

model_name = 'Sotelo b.'
ppml_sotelo_model_rta_with_domYearFE_results <-
  format_regression_table(ppml_sotelo_model_rta_with_domYearFE_results, column_name = model_name)
ppml_sotelo_model_rta_with_domYearFE_results <- rbind(ppml_sotelo_model_rta_with_domYearFE_results,
                                                      c("Number of observations",length(ppml_sotelo_model_rta_with_domYearFE$fitted.values)))
ppml_sotelo_model_rta_with_domYearFE_results <- rbind(ppml_sotelo_model_rta_with_domYearFE_results,
                                                      c("dom*year FEs","Yes"))

#Combining results
combined_results <-merge(ols_model_rta_results, ppml_model_rta_results,
                        by = "term", all = TRUE)

```

```

combined_results <-merge(combined_results, ppml_sotelo_model_rta_results,
                        by = "term", all = TRUE)
combined_results <-merge(combined_results, ols_model_rta_with_domYearFE_results,
                        by = "term", all = TRUE)
combined_results <-merge(combined_results, ppml_model_rta_with_domYearFE_results,
                        by = "term", all = TRUE)
combined_results <-merge(combined_results, ppml_sotelo_model_rta_with_domYearFE_results,
                        by = "term", all = TRUE)

desired_order <- c("RTA", "dom*year FEs", "Number of observations")

combined_results$term <- factor(combined_results$term, levels = desired_order)
combined_results <- combined_results[order(combined_results$term), ]
rownames(combined_results) <- NULL

kable(combined_results,
      caption = "Gravity Models Panel Estimations",
      format = "markdown")

```

Table 5: Gravity Models Panel Estimations

term	OLS a.	PPML a.	Sotelo a.	OLS b.	PPML b.	Sotelo b.
RTA	2.1e-01 (4.1e-02) ***	5.5e-01 (9.7e-02) ***	3.6e-01 (5.6e-02) ***	1.9e-01 (4.1e-02) ***	1.9e-01 (4.1e-02) ***	1.9e-01 (4.1e-02) ***
dom*year FEs	No	No	No	Yes	Yes	Yes
Number of observations	91506	99708	99708	91506	99708	99981

c. Comment on the difference between 4.b and 4.a and explain why there is this difference.

For question 4.a., we can notice that the domestic dummy is colinear to the importer-exporter fixed effect and is therefore absorbed (just like distance, contiguity, common language and colonial ties).

We can see that the effect of RTA on trade (or import shares) is positive and significant for all estimators (with or without domestic*year fixed effects), but that the inclusion of domestic*year fixed effects slightly reduces the coefficient.

The domestic*year fixed effect controls for general annual trends in domestic trade, without differentiation between countries. The slight drop in the RTA coefficient means that certain fluctuations specific to domestic trade were previously confused with the effect of RTA, and by controlling for them, we refine our estimate of RTA's impact on trade (or import shares). If, for instance, a global event in a given year negatively impacts domestic trade across countries (or, conversely, leads to a general increase in it), this common effect will be absorbed and removed from our estimate.

Therefore, the introduction of this fixed effect leads to a more accurate estimate of the effect of trade agreements on trade (or import shares).

d. Comment on the difference between 4.b and 3.b-3.d and explain why there is this difference.

The models from question 3 only concern the year 1990. Therefore, we only compare the effect of RTA on trade (or import shares) based on the variation between country pairs with an RTA and those without (controlling for exporter-specific and importer-specific factors, but not dyad-specific factors). As a result, we

cannot capture the time-varying effect (over multiple years) of joining an RTA for each dyad while controlling for each dyad's unique characteristics, as we do in question 4.

More specifically, we find a negative and significant estimate for RTA in model 3.b, which would not make much sense if we interpreted it as the effect of joining an RTA (with a time-varying effect). Indeed, such agreements are expected to enhance trade, not reduce it. This might be linked to the fact that such agreements only began to emerge in 1989 (at least in our dataset), meaning that most country pairs with an RTA in 1990 had just started to join them. A potential interpretation might be that these countries chose to establish an RTA precisely because their trade was relatively low compared to their other partners, and they wanted to increase it (which could explain the negative relationship between RTA and trade at the onset of joining in 1990).

However, this negative estimate does not hold in models 3.c and 3.d (where we find positive estimates, quite significant for model d), which raises doubts about the previous interpretation.

In the models from question 4, we find that RTA indeed enhanced trade, as shown by the positive and significant RTA estimates across each model in this question.

5. Now estimate the dynamic effect of joining an RTA, using an event study approach For this, you will need to define dummies for each year before joining an RTA, and dummies for the first, second, etc., year after joining an RTA. Study the pattern of these dummies. In doing so, for simplicity, use only a model in logs. Include the regression exporter*importer FEs, exporter*year FEs, importer*year FEs, and domestic trade*year FEs. Do this three times, using the following estimators: OLS, Sun & Abraham (2021) and Borusyak, Jaravel & Spiess (2023)

```
# Charger les données
# Générer les variables nécessaires
data_final <- data %>%
  filter(trade!=0) %>%
  mutate(dom = as.integer(importer == exporter),
         m = trade,
         lm = log(m)) %>%
  group_by(importer, year) %>%
  mutate(mT = sum(m),
         msh = m / mT) %>%
  ungroup() %>%
  mutate(importerXyear = as.integer(interaction(importer, year)),
         exporterXyear = as.integer(interaction(exporter, year)),
         domXyear = as.integer(interaction(dom, year)))

# Définir les indicateurs d'année avant et après RTA
# Création de la variable d'année de début de RTA
data_final <- data_final %>%
  arrange(importer, exporter, year) %>%
  group_by(importer, exporter) %>%
  mutate(RTA1 = as.integer(RTA == 1 & lag(RTA, default = 0) == 0),
         RTAstartyear1 = ifelse(RTA1 == 1, year, NA)) %>%
  mutate(RTAstartyear = mean(RTAstartyear1, na.rm = TRUE)) %>%
  ungroup() %>%
  select(-RTAstartyear1, -RTA1)

# Création des variables pour les années avant et après l'entrée en RTA
for (t in 1:10) {
  data_final <- data_final %>%
    mutate(!paste0("Brta_", t) := as.integer(year == (RTAstartyear - t)))
}
```

```

for (t in 0:10) {
  data_final <- data_final %>%
    mutate(!paste0("Arta_", t) := as.integer(year == (RTAstartyear + t)))
}

# Ajustements spécifiques pour Brta_1, Brta_10 et Arta_10
data_final <- data_final %>%
  mutate(Brta_1 = 0,
    Brta_10 = as.integer(year <= (RTAstartyear - 10)),
    Arta_10 = as.integer(year >= (RTAstartyear + 10)),
    lastcohort = as.integer(is.na(RTAstartyear)))

# Estimation OLS avec effets fixes
ols_model <- feols(lm ~ Brta_10 + Brta_9 + Brta_8 + Brta_7 + Brta_6 + Brta_5 +
  Brta_4 + Brta_3 + Brta_2 + Brta_1 +
  Arta_0 + Arta_1 + Arta_2 + Arta_3 + Arta_4 + Arta_5 + Arta_6
  + Arta_7 + Arta_8 + Arta_9 + Arta_10 |
  importerXyear + exporterXyear + domXyear + pair_id,
  cluster = ~pair_id, data = data_final)

# Estimation Sun & Abraham avec l'approche "event study"
sun_abraham_model <- feols(lm ~ sunab(RTAstartyear, year) | importerXyear
  + exporterXyear + domXyear + pair_id,
  cluster = ~pair_id, data = data_final)

# Estimation Borusyak, Jaravel & Spiess (DID Imputation)
first_stage <- feols(lm ~ 1 | importerXyear + exporterXyear + domXyear
  + pair_id, data = data_final)

data_final$residuals_first_stage <- resid(first_stage)

bjs_model <- did_imputation(yname = "residuals_first_stage", idname = "pair_id",
  tname = "year", gname = "RTAstartyear",
  pretrends = 10, horizon = -10:10, data = data_final,
  cluster_var = "pair_id")

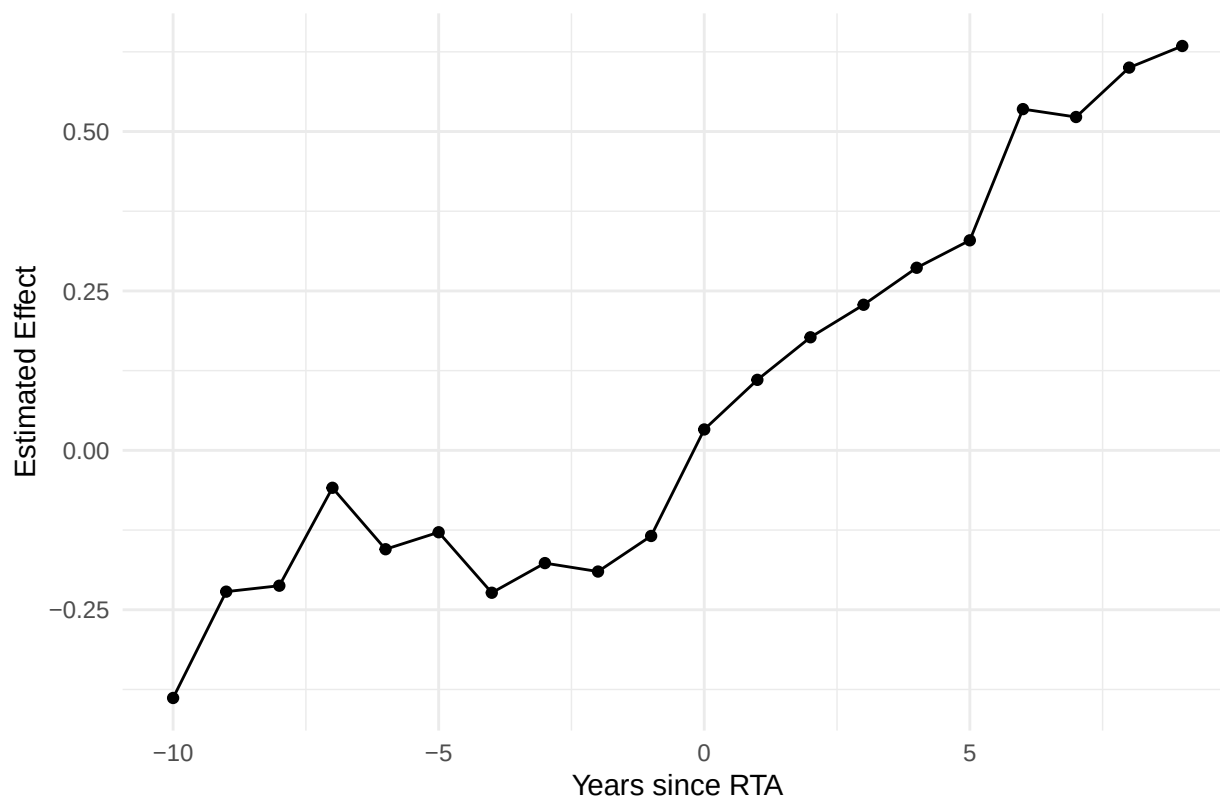
# Préparation des données pour les graphiques
# Extraire les coefficients pour chaque modèle
ols_estimates <- coef(ols_model)
sun_abraham_estimates <- coef(sun_abraham_model)
bjs_estimates <- bjs_model$estimate

# Graphique des résultats pour OLS
ols_plot_data <- data.frame(
  Time = -10:9,
  Estimate = ols_estimates[grep("Brta_|Arta_", names(ols_estimates))]
)

ggplot(ols_plot_data, aes(x = Time, y = Estimate)) +
  geom_point() +
  geom_line() +
  labs(title = "OLS Event Study for RTA", x = "Years since RTA", y = "Estimated Effect") +
  theme_minimal()

```

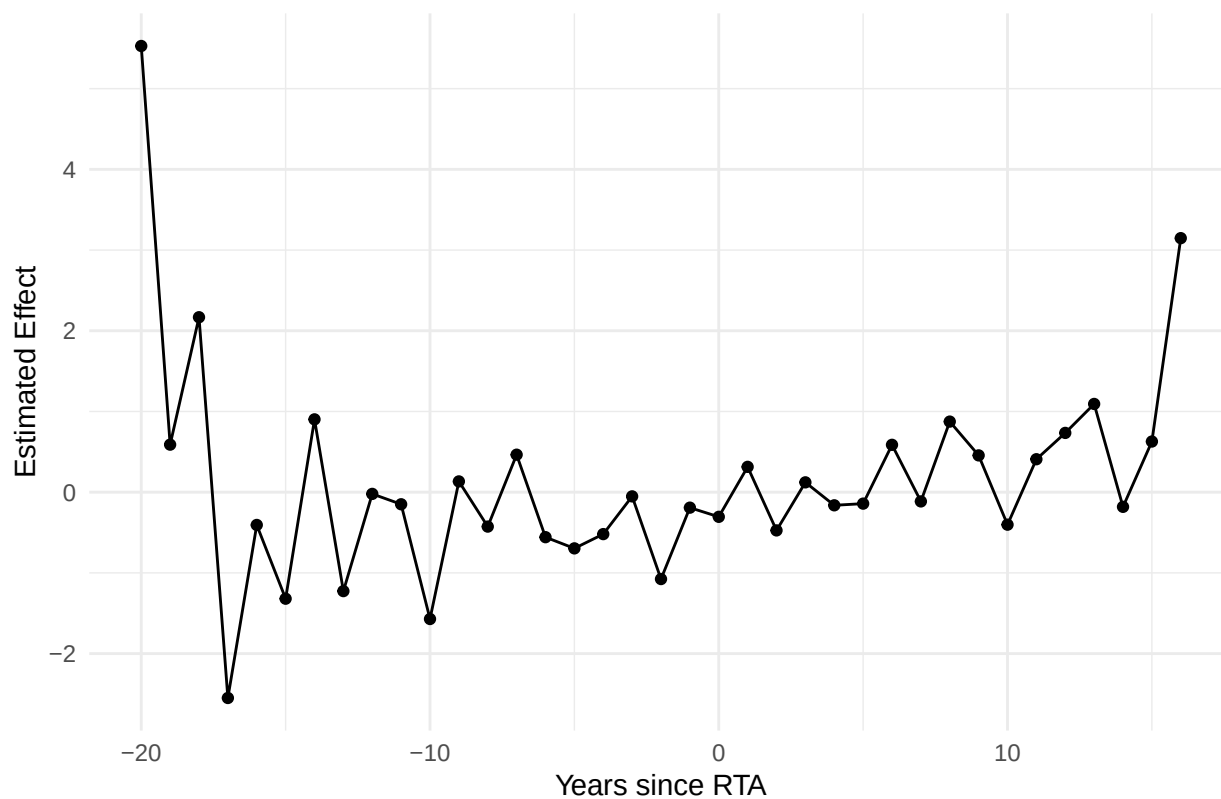
OLS Event Study for RTA



```
# Graphique des résultats pour Sun & Abraham
sa_plot_data <- data.frame(
  Time = -20:16,
  Estimate = sun_abraham_estimates
)

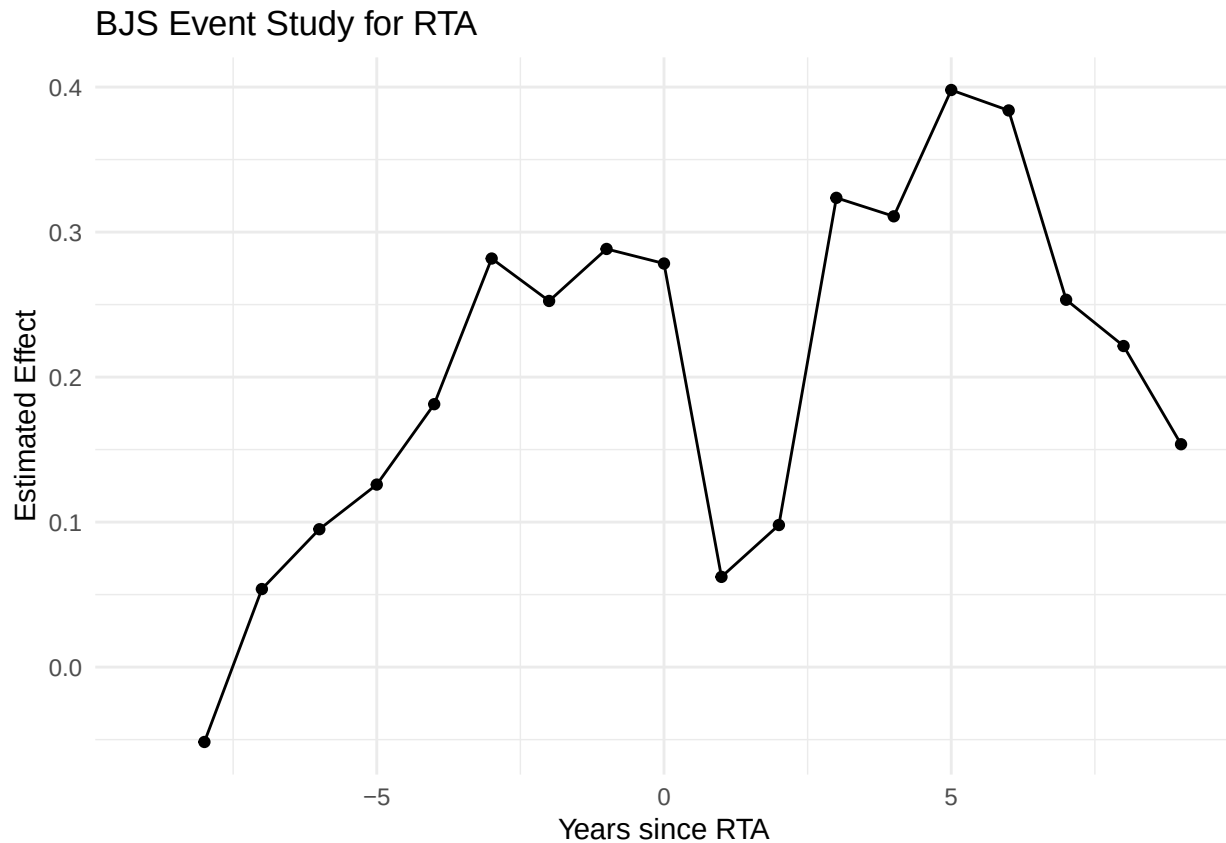
ggplot(sa_plot_data, aes(x = Time, y = Estimate)) +
  geom_point() +
  geom_line() +
  labs(title = "Sun & Abraham Event Study for RTA", x = "Years since RTA", y = "Estimated Effect") +
  theme_minimal()
```

Sun & Abraham Event Study for RTA



```
# Graphique des résultats pour BJS
bjs_plot_data <- data.frame(
  Time = -9:9,
  Estimate = bjs_estimates
)

ggplot(bjs_plot_data, aes(x = Time, y = Estimate)) +
  geom_point() +
  geom_line() +
  labs(title = "BJS Event Study for RTA", x = "Years since RTA",
    y = "Estimated Effect") +
  theme_minimal()
```



Comment on the differences between the results from these estimators.

As the packages `eventstudyinteract` and `did_imputation` have slightly different parameter sets in R compared to Stata, there might be some model specifications provided in the Stata code that are missing in my R script, as I used ChatGPT for the translation. Therefore, the following results should be interpreted with caution.

We can see that with the OLS model, the estimated effect of RTA on trade increases from 10 years before joining the RTA until 10 years after joining. The effect during the years before joining does not fully increase, but at least shows an upward trend. The effect post-RTA is clearly positive and increasing, in line with our expectations. We should note that trade increased even before joining, perhaps partly due to an announcement effect.

The Sun & Abraham model provides mixed evidence. The post-RTA trend is positive and increasing, but not as much as with the OLS model, and shows some fluctuations. It shows a significant drop in trade 20 years before joining the RTA, which might indicate some computational issues in the model estimation (as this pattern doesn't seem entirely logical).

Finally, the BJS model shows a positive effect of RTA on trade, reaching a maximum 5 years after joining. However, this is preceded by a large drop immediately after joining the RTA (which doesn't entirely make sense). This would suggest that it takes time for the RTA to have its maximum impact on trade (several years). Furthermore, the positive effect of RTA decreases after 5 years, which would suggest that this increase could be reversible at some point (which seems unusual).

We conclude that, with the data and R script used, the OLS model might be the one that best fits what we would expect in reality.